Activity 7 Correlation Detection

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Activity 7

Aim

To analyze relationships among seven dairy farm variables using a correlation matrix, scatter plots, quadratic regression, outlier detection, and partial correlation in R.

Task 1 — Correlation matrix

Code:

```
set.seed(109143)
n <- 14
rainfall <- round(rnorm(n, 118.4, 39.1), 1)
grass_height <- round(34.5 + 8.8 * (
  0.91 * (rainfall - 118.4) / 39.1 +
    0.41 * rnorm(n)), 1)
milk_yield <- round(13.8 + 2.9 * (
  0.85 * (rainfall - 118.4) / 39.1 +
    0.53 * rnorm(n)), 1)
fert_dev <- round(
  seq(-28.0, 28.0, length.out = n)[order(runif(n))], 1)
crop_damage <- round(
```

```
  5.9 + 23.5 * (fert_dev / 28.0)^2 +
    rnorm(n, 0, 1.3), 1)
tractor_hours <- round(rnorm(n, 89, 19.6), 0)
fuel_cost <- round(44.4 + 9.8 * (
  0.13 * (tractor_hours - 89) / 19.6 +
    0.99 * rnorm(n)), 1)
tractor_hours[11] <- 153
fuel_cost[11] <- 74.7

case <- data.frame(
  rainfall, grass_height, milk_yield, fert_dev,
  crop_damage, tractor_hours, fuel_cost
)

# Correlation matrix
cor_matrix <- cor(case)
print(round(cor_matrix, 2))

# Number of distinct variable pairs
cat("Observations:", nrow(case), "\n")
cat("Variables:", ncol(case), "\n")
cat("Distinct pairs:", choose(ncol(case), 2), "\n")

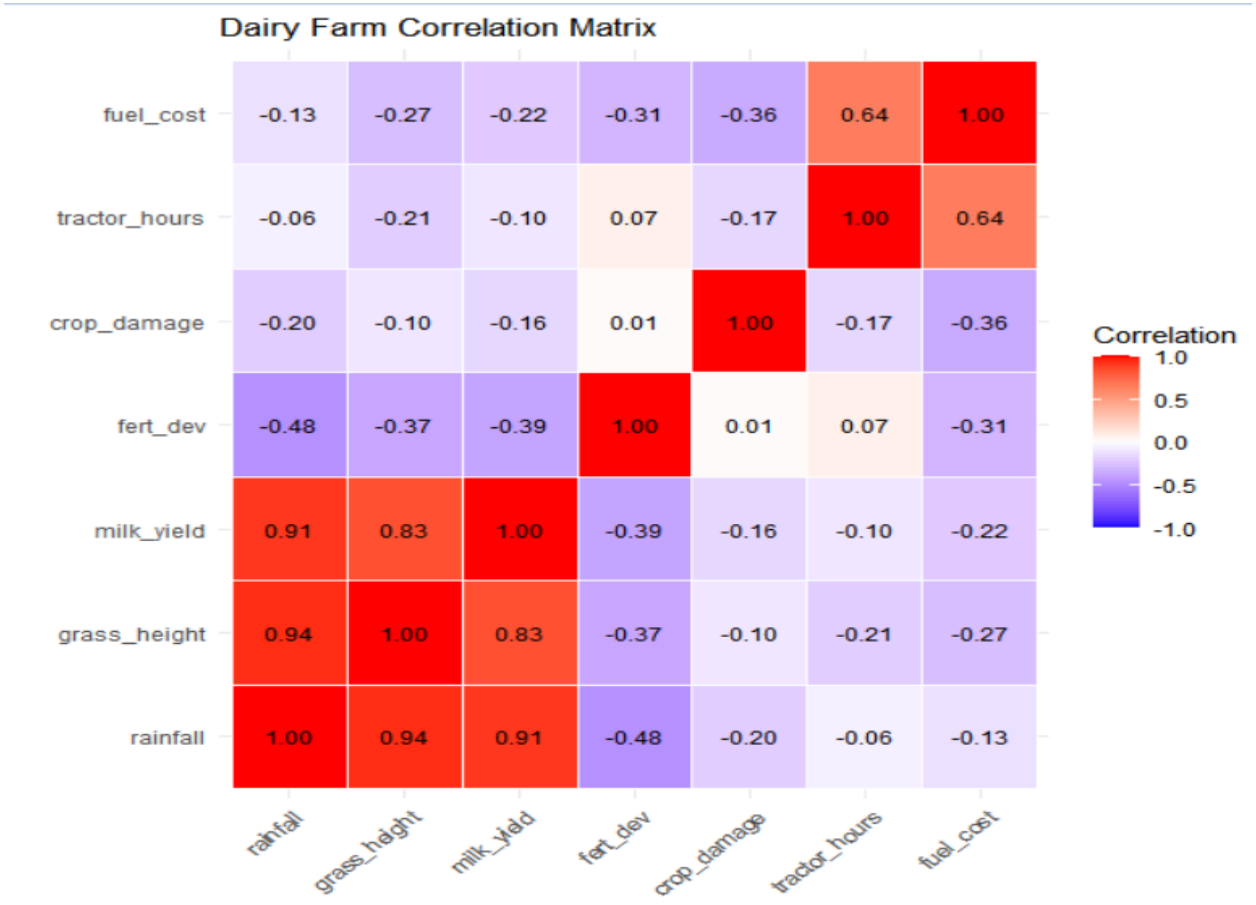
# Correlogram using ggplot2
library(ggplot2)

cor_long <- as.data.frame(as.table(cor_matrix))
names(cor_long) <- c("Variable1", "Variable2", "Correlation")

ggplot(cor_long,
```

```
    aes(x = Variable1, y = Variable2, fill = Correlation)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = sprintf("%.2f", Correlation)), size = 3) +
  scale_fill_gradient2(
    low = "blue", mid = "white", high = "red",
    midpoint = 0, limits = c(-1, 1)
  ) +
  labs(title = "Dairy Farm Correlation Matrix",
       x = "", y = "", fill = "Correlation") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Output



Answer:

- Number of observations: $n = 14$
- Number of variables: 7
- Number of distinct variable pairs:

\[\frac{7(7-1)}{2}=\boxed{21} \]

Task 2 — Find the strongest clues

Use the off-diagonal values of the correlation matrix to find the strongest positive and negative pairs.

Code:

```
# Find distinct pairs only; exclude diagonal and duplicate pairs
upper <- upper.tri(cor_matrix)
pair_values <- cor_matrix[upper]

strongest_positive <- which.max(pair_values)
strongest_negative <- which.min(pair_values)

pairs <- which(upper, arr.ind = TRUE)

cat("Strongest positive pair:",
   rownames(cor_matrix)[pairs[strongest_positive, 1]], "and",
   colnames(cor_matrix)[pairs[strongest_positive, 2]], "\n")
cat("Correlation:",
   round(pair_values[strongest_positive], 2), "\n")

cat("Strongest negative pair:",
```

```
    rownames(cor_matrix)[pairs[strongest_negative, 1]], "and",
    colnames(cor_matrix)[pairs[strongest_negative, 2]], "\n")
cat("Correlation:",
    round(pair_values[strongest_negative], 2), "\n")
```

Output

```
> cat("Strongest negative pair:",
+     rownames(cor_matrix)[pairs[strongest_negative, 1]], "and",
+     colnames(cor_matrix)[pairs[strongest_negative, 2]], "\n")
Strongest negative pair: rainfall and fert_dev
> cat("Correlation:",
+     round(pair_values[strongest_negative], 2), "\n")
Correlation: -0.48
>
```

Strength classification:

Strength classification:

Absolute correlation, $\|r\|$	Classification
Below 0.30	Weak
0.30 to 0.70	Moderate
Above 0.70	Strong

Answer: Copy the two variable pairs and their correlation values from the R output. Classify each using the table above.

Task 3 — The deceptive zero

Code:

```
# Plot crop damage against fertilizer deviation
plot(case$fert_dev, case$crop_damage,
     pch = 19, col = "blue",
     main = "Crop Damage vs Fertilizer Deviation",
     xlab = "Fertilizer Deviation (kg per ha)",
     ylab = "Crop Damage (%)")

# Straight-line fit
linear_fit <- lm(crop_damage ~ fert_dev, data = case)
abline(linear_fit, col = "red", lwd = 2)

# Quadratic fit
quadratic_fit <- lm(
  crop_damage ~ fert_dev + I(fert_dev^2),
  data = case
)

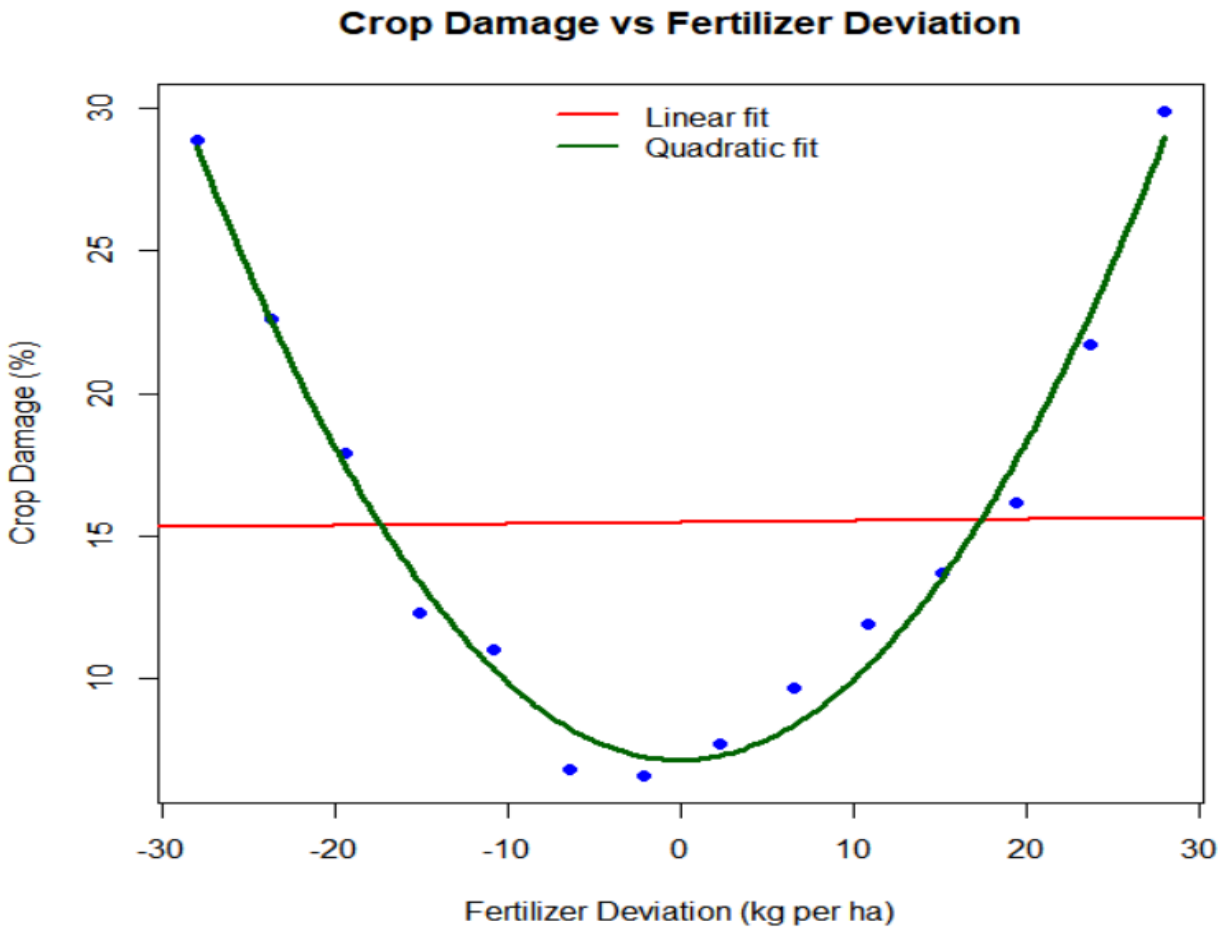
x_grid <- seq(min(case$fert_dev), max(case$fert_dev),
              length.out = 200)
y_grid <- predict(quadratic_fit,
                  newdata = data.frame(fert_dev = x_grid))
lines(x_grid, y_grid, col = "darkgreen", lwd = 3)

# Correlation and quadratic R-squared
r <- cor(case$fert_dev, case$crop_damage)
r_squared <- summary(quadratic_fit)$r.squared
```

```
cat("Pearson correlation r:", round(r, 3), "\n")
cat("Quadratic R-squared:", round(r_squared, 3), "\n")

legend("top",
       legend = c("Linear fit", "Quadratic fit"),
       col = c("red", "darkgreen"),
       lwd = 2, bty = "n")
```

Output



Answer and interpretation:

- Pearson correlation r: report the value printed by R.
- Quadratic model R^2: report the value printed by R.

Why can correlation miss this relationship? Fertilizer deviation and crop damage can have a curved, U-shaped relationship, so positive and negative linear trends on opposite sides can cancel out and make Pearson's correlation close to zero even when the quadratic relationship is strong.

Task 4 — The outlier alibi

The evidence states that row 11 was entered by a trainee. Check whether it is an outlier in the relationship between tractor hours and fuel cost.

Code:

```
# Scatter plot of all observations
plot(case$tractor_hours$, $case$fuel_cost,
    pch = 19, col = "blue",
    main = "Fuel Cost vs Tractor Hours",
    xlab = "Tractor Hours",
    ylab = "Fuel Cost (thousand rupees)")

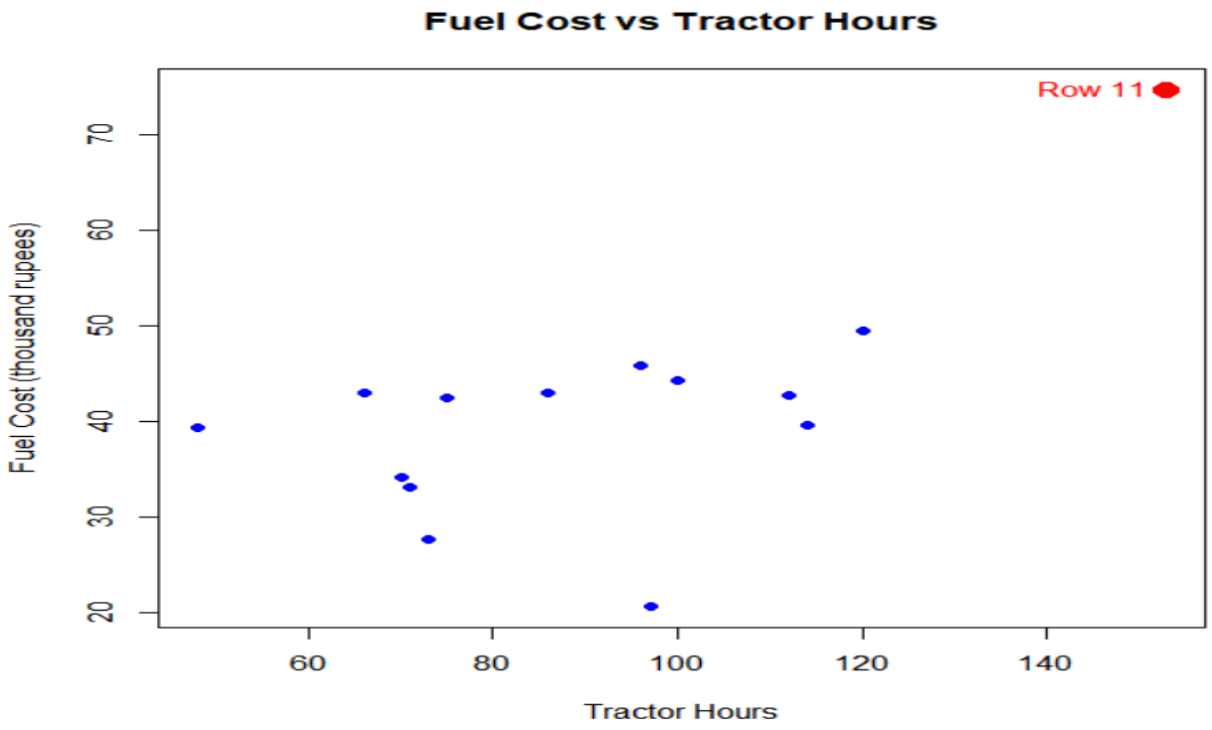
# Highlight row 11
points(case$tractor_hours$[11], $case$fuel_cost[11],
      pch = 19, col = "red", cex = 1.8)
text(case$tractor_hours$[11], $case$fuel_cost[11],
    labels = "Row 11", pos = 2, col = "red")
```

```
# Correlation with all observations
r_all <- cor(case$tractor_hours, case$fuel_cost)

# Correlation excluding row 11
case_clean <- case[-11, ]
r_without <- cor(case_clean$tractor_hours, case_clean$fuel_cost)

cat("Suspicious row: 11\n")
cat("Correlation with all rows:", round(r_all, 3), "\n")
cat("Correlation without row 11:", round(r_without, 3), "\n")
```

Output



Answer:

- Suspicious row: 11
- Correlation with all 14 observations: report `r_all`.
- Correlation without row 11 (13 observations): report `r_without`.

Interpretation: The unusual observation at row 11 can substantially change the correlation coefficient. Compare the two values to determine whether it inflates or reduces the apparent linear relationship.

Task 5 — The prime suspect

Calculate the correlation between grass height and milk yield, then check how both variables relate to rainfall.

Code:

```
# Simple correlations
r_grass_milk <- cor(case$grass_height, case$milk_yield)
r_rain_grass <- cor(case$rainfall, case$grass_height)
r_rain_milk <- cor(case$rainfall, case$milk_yield)

# Partial correlation controlling for rainfall
grass_resid <- resid(lm(grass_height ~ rainfall, data = case))
milk_resid <- resid(lm(milk_yield ~ rainfall, data = case))

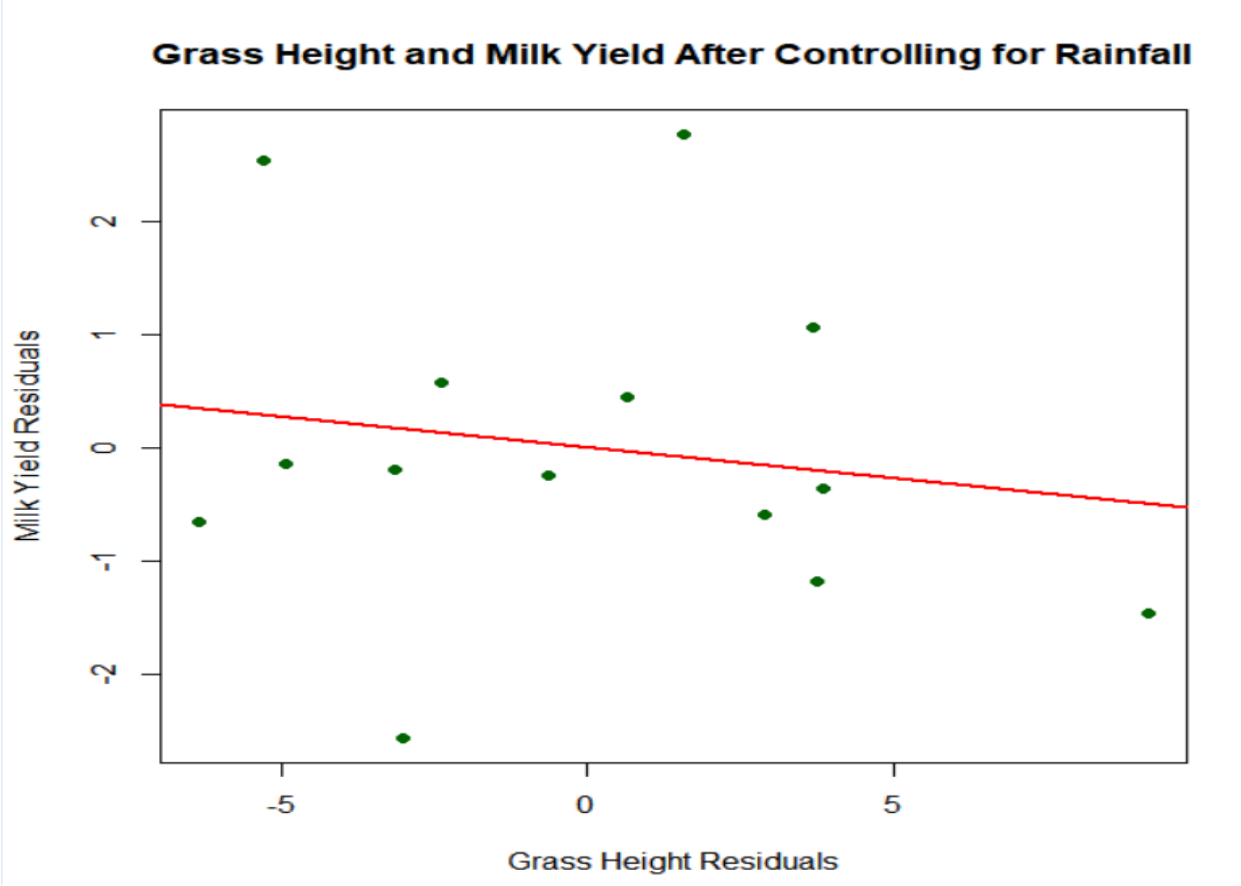
partial_r <- cor(grass_resid, milk_resid)

cat("Correlation: grass height and milk yield:",
   round(r_grass_milk, 3), "\n")
cat("Correlation: rainfall and grass height:",
   round(r_rain_grass, 3), "\n")
```

```
cat("Correlation: rainfall and milk yield:",
    round(r_rain_milk, 3), "\n")
cat("Partial correlation controlling for rainfall:",
    round(partial_r, 3), "\n")

# Optional residual scatter plot
plot(grass_resid, milk_resid,
     pch = 19, col = "darkgreen",
     main = "Grass Height and Milk Yield After Controlling for Rainfall",
     xlab = "Grass Height Residuals",
     ylab = "Milk Yield Residuals")
abline(lm(milk_resid ~ grass_resid), col = "red", lwd = 2)
```

Output



Answer:

- $r(grassheight, milkyield)$: report the first value.
- $r(rainfall, grassheight)$: report the second value.
- $r(rainfall, milkyield)$: report the third value.
- Partial correlation controlling for rainfall: report `partial_r`.

Two-sentence verdict:

Grass height does not necessarily cause higher milk yield because rainfall may influence both grass growth and milk production, and correlation alone cannot prove causation. A randomized controlled feeding experiment, in which cows are assigned to different forage treatments while other conditions are controlled, could help test whether forage quality or availability causes changes in milk yield.

Final conclusion

Correlation analysis helps identify strong associations, nonlinear relationships, and the effect of outliers. For this dairy farm, the manager should examine the quadratic fertilizer–crop damage relationship, investigate the suspicious tractor and fuel record, and avoid interpreting the grass–milk association as proof of cause.

Code of Conduct: I certify that this work is my own.

Signature: ***Udhayakumar***